



Marco Barca

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Date of birth: 05/10/1995 **Nationality**: Italian

WORK EXPERIENCE

NPO Torino

City: Torino | **Country**: Italy

[20/02/2024 – Current]

Cloud Solutions Engineer

- Designed and implemented an end-to-end ML pipeline for IT ticket intelligence at a global manufacturing enterprise: raw data ingestion from ServiceNow APIs, semantic embeddings, UMAP dimensionality reduction, auto-tuned HDBSCAN unsupervised clustering, and a multi-stage LLM orchestration layer (Azure AI Foundry) for automated text normalization, topic labeling and Knowledge Base generation. Delivering interactive dashboards and business reports on thousands of incidents and service requests.
- Designed, developed and productized a multi-tenant SaaS platform that fully automates telephone survey workflows from orchestrating outbound calls to delivering AI-generated business reports with no human intervention. Built on a serverless, event-driven architecture (Azure Functions, Azure Communication Services), with Azure Speech for call transcription, Azure AI Foundry and LangChain for the report generation pipeline, a RAG-based conversational agent (Azure Cognitive Search + semantic embeddings) enabling stakeholders to query and explore generated reports in natural language, and native multilingual support (Azure Translator + neural TTS) to operate across different languages and regions deployed via Azure Pipelines CI/CD with multi-tenant isolation via Azure Entra ID.
- Designed and delivered data pipeline and lakehouse architectures across multiple engagements, including ERP integrations (Zucchetti REST API, OAuth2), Medallion architecture implementations on PostgreSQL, and data processing workflows with PySpark and Microsoft Fabric, ensuring data consistency, fault tolerance and traceability for downstream BI reporting.

[01/01/2025 – Current]

Innovation Team lead

- Led pre-sales and solution design for strategic clients: requirements analysis, high-level architecture, effort estimation, and PoC delivery.
- Driving internal LLM adoption across engineering teams: introducing agentic workflows, AI-assisted development tooling, and organizing hands-on workshops to upskill developers.
- Defined the technical direction for innovation initiatives, evaluating emerging technologies and aligning solutions with business objectives.

V3 Advisory

City: Milano | **Country**: Italy

[01/01/2026 – Current]

Solution Architect & Tech Advisor

- Solution Architect responsible for technical governance and end-to-end architecture of cloud-based and AI-driven solutions. - Driving system design, architectural decisions, and alignment between business objectives and scalable technical implementations.

EDUCATION AND TRAINING

[2018 – 2020]

Bachelor's Degree in Computer Engineering

Università degli studi di Salerno

City: Salerno | **Country**: Italy | **Thesis**: Experimental Characterization of a System for Ethnicity Recognition under Occluded Conditions

The main purpose of this work is to test the accuracy of Computer Vision systems on different image datasets and their masked counterparts in order to calculate the error in prediction accuracy.

[2020 – 2023]

Master's Degree in Computer Engineering - Cybersecurity

Politecnico di Torino

City: Torino | **Country:** Italy | **Final grade:** 103 | **Thesis:** Modeling and Analyzing Attacker Behavior through Graph Construction

The primary aim of this study is to model and analyze computer attacks stemming from Capture The Flag (CTF) competitions, utilizing Generative AI models. This is achieved through the construction of graphs that delineate the steps involved, emphasizing the techniques employed and the Digital Artefacts produced. These artifacts encompass data, files, or elements generated as a consequence of human actions, processes, or interactions within a digital environment.

[01/2022 – 07/2022]

Master in Blockchain & Digital Assets

MasterZ

[01/07/2022 – 31/07/2022]

2°place - Encode x Algorand Hackathon

Encode <https://www.encode.club>

A 4-week hackathon focused on the Algorand blockchain. I achieved second place by developing a decentralized crowdfunding platform.

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING B2 **READING** B2 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Blockchain | Cybersecurity | Backend Development | Frontend Development | Machine Learning | Computer Vision | Generative AI | Data Analytics & Visualization | Hadoop | Spark | PL/SQL | SQL | Docker | Spring | C/C++ | JavaScript | Kotlin | Python | Node.js | Java | Microsoft Fabric | Microsoft Azure